

Practice Quiz: Creative Perception

Part A: Multiple Choice

1. In Active Inference, creative perception is best described as: A) Having sharper senses than other people B) Detecting prediction errors that others' models suppress or fail to generate C) Gathering more data than competitors through better instrumentation D) Thinking more carefully about available information
2. Functional fixedness arises in Active Inference when: A) The agent lacks knowledge about a problem domain B) The agent has excessive precision on categorical priors about object functions C) The agent has too few prior beliefs to make predictions D) The environment lacks sufficient affordances for creative action
3. The Generic Parts Technique overcomes functional fixedness by: A) Adding new information about the object's history B) Removing functional labels and describing objects purely by physical properties C) Asking the inventor to collaborate with a team D) Increasing the time available for problem-solving
4. George de Mestral invented Velcro because his generative model: A) Was narrowly focused on textile manufacturing B) Included knowledge of fastening mechanisms, allowing him to perceive the burr's hook-and-loop structure as a solution C) Was devoid of prior beliefs about fastening, leaving him open to any possibility D) Was identical to other inventors who also noticed burrs but acted faster
5. Latent needs are needs that users: A) Have clearly articulated in market research surveys B) Do not actually have but that inventors imagine for them C) Have but cannot articulate because their model has adapted to the status quo D) Will develop in the future but do not currently experience
6. The Shinkansen bullet train nose redesign based on kingfisher beak morphology is an example of: A) Random trial and error in aerodynamic design B) Cross-domain pattern recognition — transferring a structural principle from ornithology to engineering C)

Copying a natural form without understanding the underlying principle D) Standard computational fluid dynamics optimization

7. The "inversion" reframing technique works by: A) Physically turning the object upside down B) Asking how to make the problem worse, then inverting each answer into a potential solution C) Reversing the order of steps in a manufacturing process D) Consulting experts from the opposite end of the political spectrum

Part B: Short Answer / Analysis

1. A product designer at a kitchenware company is asked to improve their company's vegetable peeler. Every proposal they generate is a variation on the existing peeler design — different handle shapes, different blade angles, different materials. Using the concept of functional fixedness and Active Inference, diagnose why they are stuck. Then propose two specific perceptual reframing techniques they could apply to break out of this pattern, explaining what each technique changes about their generative model.
2. Akio Morita of Sony perceived the need for a personal portable music player (the Walkman) before any consumer had articulated this need. Analyze this as latent need detection using Active Inference concepts. What was the prediction error in Morita's model of user behavior? Why could he detect this need while consumers themselves could not? What does this tell us about the relationship between the inventor's generative model and the user's generative model?
3. You are an inventor working on a solution for urban food delivery. Apply three different perceptual reframing techniques from this module to your problem: (a) cross-domain pattern recognition — find a structural analogy from biology, (b) scale shifting — describe the problem at the city level versus the individual meal level, and (c) stakeholder rotation — describe the problem from the restaurant's perspective versus the customer's. For each reframe, identify one new insight about the problem that was not visible from your original perspective.